STRAIT OF HORMUZ CRISIS SIMULATION

Scenario Document — Operation Epic Fury

Document Type: Complete scenario definition for agent-based geopolitical simulation **Conflict start:** ~February 25, 2026 (Day 1) **Scenario snapshot:** Day 18 of hostilities (March 15, 2026) **Data sources:** IranMonitor.org (OSINT dashboard), public reporting, IRGC/IDF statements, Polymarket prediction markets, GeoConfirmed satellite/OSINT, MarineTraffic, Cloudflare/IODA internet monitoring

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PART I: CONFLICT OVERVIEW

Origins

The conflict began when Israel conducted a decapitation strike against Iran, killing Supreme Leader Ali Khamenei on Day 1. The strike was part of a broader Israeli-US military operation — designated **Operation Epic Fury** (US) and **Operation Roaring Lion** (Israel) — aimed at permanently degrading Iran's nuclear program, missile capability, and military infrastructure.

The strike did not occur in a vacuum. Since late 2025, Iran had been experiencing its largest domestic uprising in the history of the Islamic Republic, with over 60,500 verified killed, 11,000+ injured, and 42,000+ detained by security forces. The IRGC was already engaged in a brutal internal crackdown when the external war began, forcing it to fight on two fronts simultaneously.

The operation rapidly escalated from targeted Israeli strikes into a regional war involving direct Iranian retaliation against multiple Gulf states, proxy activation across the Middle East, and a global energy crisis centered on the Strait of Hormuz.

War Aims (Stated)

Actor	Stated Objective
Israel	Regime change in Iran. Permanent elimination of nuclear and missile threat.
United States	Officially regime change, but President Trump's statements oscillate between total victory rhetoric and openness to negotiation. "The terms aren't good enough yet" (March 14) implies active back-channel discussions.
Iran (IRGC)	Survival of the Islamic Republic. Publicly denying all ceasefire proposals. Framing the war as existential resistance. Iran FM has however acknowledged war-ending conditions publicly, creating a gap between military and diplomatic postures.

Timeline Summary (Days 1-18)

Day 1 (~Feb 25): Israeli strike kills Supreme Leader Ali Khamenei. Coordinated US-Israeli air campaign begins. Iranian air force and naval surface fleet targeted in opening hours. Mojtaba Khamenei reportedly designated as successor.

Days 1-3: Iran's conventional air force and most naval surface vessels destroyed. IRGC activates asymmetric doctrine — ballistic missiles, drones, fast boats, coastal defense missiles, and mines. Proxy network receives activation orders. Mojtaba Khamenei's location and status become uncertain.

Days 3-7: IRGC begins missile and drone strikes on Gulf states. First tanker attacks in the Strait of Hormuz. Kata'ib Hezbollah and Iraqi PMF begin attacks on US forces and embassy in Iraq. Oil markets spike.

Days 7-14: Iran implements selective transit regime in the Strait of Hormuz — allowing Chinese-flagged, Indian-flagged, and select other vessels to transit while targeting others. At least 10 tankers struck. Insurance markets freeze for non-permitted flag states. Iran strikes Dubai, Abu Dhabi, and Omani targets. US embassy in Baghdad hit, radar system destroyed. Oil reaches $98/barrel (up 56% from pre-war ~$63).

Days 14-18 (current): Escalation widens dramatically. Iran launches combined ballistic missile and drone assaults on UAE, Bahrain, and Kuwait alongside continued strikes on Israel. Iran issues port evacuation warnings for Abu Dhabi, Dubai, and Fujairah — telegraphing further attacks. Iran deploys Sejjil IRBM and Fattah hypersonic missiles for the first time in this war. Israel announces "wide-scale" new wave of strikes on western Iran, including heavy pre-dawn explosions at Isfahan and Vahdat Air Base in Dezful. Iran Space Research Centre struck in western Tehran. IDF issues evacuation alert for Eastern Tabriz. US strikes destroy military infrastructure on Kharg Island (Trump: "totally demolished"; oil export terminal status unconfirmed — commercial satellite imagery delayed 3 days). Three Iranian missiles fired at a Turkish air base; all intercepted. Iran's FM publicly acknowledges war-ending conditions while official channels deny ceasefire talks. Ukrainian President Zelensky alleges Russia is supplying Iran with Shahed drones for use against US and Israeli targets.

Critical concurrent event: The Iranian domestic uprising that began in late 2025 continues during the war. Over 60,500 verified killed, 42,486+ detained. IRGC conducting simultaneous external war and internal crackdown on its own population.

PART II: CURRENT STATE OF PLAY (DAY 18)

Iranian Military Capabilities

Capability	Status
Supreme Leader (Khamenei)	Dead (Day 1)
Successor (Mojtaba Khamenei)	Status disputed. Trump claims dead. Iran FM denies. Unconfirmed.
Conventional air force	Destroyed
Naval surface fleet	Destroyed
Civilian government	Sidelined. IRGC in de facto control of war operations. FM operating diplomatic track that may or may not have IRGC authorization.
IRGC ground forces	Degraded. IDF reports 6,000+ IRGC killed, 15,000+ wounded (Operation Roaring Lion figures — may be inflated).
Ballistic missile arsenal	Active and deploying advanced systems: Sejjil IRBM (newly introduced this war), Fattah hypersonic missile. Stocks depleting but Iran introducing its most capable weapons, not just legacy systems.
Drone capability	Active. Ongoing production in hardened/underground facilities. Possibly being resupplied by Russia with Shahed drones (Zelensky allegation, unconfirmed).
Asymmetric naval (fast boats, mines, coastal anti-ship missiles)	Operational. IRGC has threatened to turn the Strait of Hormuz into a "trap."
Cyber capability	Believed intact. Not yet deployed at scale — assessed as a held escalation card.

Proxy network (Quds Force coordination)	Active but coordination degraded. Houthis notably showing RESTRAINT — a significant signal. KH/PMF active in Iraq.
Air defense	Largely destroyed. Iranian airspace effectively controlled by US/Israel.
Kharg Island	"Heavily struck" (confirmed). Military infrastructure destroyed. Oil terminal status uncertain (sat imagery delayed 3 days).
Internal security apparatus	Engaged in massive domestic crackdown simultaneous with external war. Live fire on protesters, mass detention, internet degradation, targeting of lawyers and medics, informant network rollups.

IRGC Organizational Degradation

The 21,000+ casualty figure (6,000 killed, 15,000 wounded per IDF claims) represents serious institutional damage:

- Command structure disrupted — senior officers killed in strikes
- Technical expertise lost — missile and drone operators, engineers
- Institutional knowledge destroyed — cannot be replaced quickly
- BUT: IRGC is a large organization (~190,000 personnel including Basij mobilization capacity). 21,000 casualties is painful but not organizationally fatal.
- The question: are casualties concentrated in critical specialties (missile crews, air defense, command) or distributed across the force?

Israeli Military Status

Domain	Status
Strike campaign	Ongoing "wide-scale" deep strikes into western Iran (Isfahan, Dezful, Tehran targets). Targeting nuclear sites, air bases, IRGC leadership, missile production, space research facilities.
Interceptor crisis	**CRITICALLY LOW on ballistic missile interceptors.** $825M emergency procurement authorized. Arrow/David's Sling stocks depleted from sustained Iranian missile campaigns. This is a critical vulnerability — Iran's continued launches are an attrition strategy against Israeli defense capacity.
Air operations	F-35s operating at extended range. Sustained high-tempo operations.
Northern border	Hezbollah conducting calibrated activity. IDF reserves deployed.
War aim	Regime change. Racing to destroy targets before any ceasefire.

Israeli interceptor depletion is a pivotal variable. Each Iranian Sejjil or Fattah that forces an Arrow/David's Sling intercept drains a finite and critically low resource. If Iran can sustain launches faster than Israel can replenish interceptors, Israel faces an air defense gap that could lead to mass casualties from missile strikes on population centers. This may be Iran's most viable path to imposing unbearable cost on Israel.

US/Coalition Military Status

Domain	Status

Air campaign	Week 3. Running out of high-value fixed targets. Shift to dispersed/mobile targets with diminishing returns. Kharg Island military infrastructure destroyed.
Naval forces	USS Abraham Lincoln carrier strike group in Persian Gulf. Leading multinational naval coalition formation.
Coalition expansion	US has requested allied naval deployment to Strait of Hormuz from UK, South Korea, and Japan. Status: under review by all three. UK considering minehunter drone deployment.
Casualties	Low but nonzero. Embassy in Baghdad hit (radar system destroyed).
Munitions	Significant expenditure of precision-guided munitions. Stocks drawing down.
Trump posture	Oscillating. "Totally demolished Kharg Island." Threatened strikes "just for fun." Demanded Iran's surrender. But also: "terms aren't good enough yet" — acknowledging negotiation track exists. Publicly questioning whether Mojtaba Khamenei is alive.

Strait of Hormuz Status

Iran is NOT conducting a total blockade. It is operating a **selective transit regime** — effectively weaponizing access to the strait:

Permitted transit:

- Chinese-flagged vessels
- Indian-flagged vessels
- Select other non-Western, non-Gulf state flags (assessed: Russian, Pakistani, others)

Targeted/blocked:

- Western-flagged vessels (US, UK, EU registry)
- Gulf state vessels (Saudi, UAE, Kuwaiti, Bahraini flag)
- Vessels assessed as serving US-allied interests regardless of flag

Iran has threatened to turn the strait into a "trap" — suggesting potential for escalation from selective blockade to active naval combat zone.

Result:

- Days-long shipping backlog outside the strait
- Massive re-flagging scramble toward Chinese/Indian registries
- Western maritime insurance suspended for Gulf transit
- Chinese state insurer (PICC) offering coverage for permitted flags
- De facto transfer of maritime security authority from US-led order to an Iranian/Chinese-mediated access regime

Oil Market

Indicator	Value
Pre-war price	~$63/barrel
Current price	$98/barrel (+56%)
Strait flow	Partial — Chinese/Indian traffic flowing; Western/Gulf blocked

Kharg Island	Heavily struck. Oil terminal status unknown (sat imagery delayed). If terminal is damaged/destroyed, Iran loses ~90% of its own oil export capacity.
Strategic reserves	US, China, Japan, EU releasing reserves
Market expectation	Pricing in ceasefire within weeks. If expectation drops, price spikes.

Why $98, not $300: The selective blockade means major importers (China, India) are still receiving oil. Strategic reserves are being drawn down. Russian oil is backfilling demand. Markets expect resolution. The $98 price is actually bad news for Iran's strategy — it means the world is absorbing the disruption.

Gulf States Damage

UAE: Dubai, Abu Dhabi struck. Iran issued port evacuation warnings for Abu Dhabi, Dubai, and Fujairah — telegraphing further attacks. Foreign worker evacuation accelerating. Economic model (business hub, tourism, expat economy) broken until war ends.

Bahrain: Struck in combined missile/drone assault. US Fifth Fleet HQ located there. Shia majority population creates internal security dimension.

Kuwait: Struck in latest wave. Oil infrastructure and US military installations targeted.

Saudi Arabia: Oil infrastructure hit. East-West pipeline at maximum throughput. Eastern Province under heightened security.

Oman: Previously hit despite traditional neutrality. Mediation role potentially compromised.

Turkey / NATO

- 3 Iranian missiles fired at Turkish air base — all intercepted. No casualties.
- Article 5 NOT invoked.
- Turkish Foreign Minister actively engaged in diplomatic outreach — positioning Turkey as potential ceasefire broker.
- Turkey playing dual role: Article 5 leverage holder AND active mediator.

Russia

- Zelensky alleges Russia supplying Shahed drones to Iran (CNN interview, March 14).
- If confirmed: Russia is a co-belligerent, not a bystander.
- Iran claims US/Israel using "copycat drones" to frame Iran — information warfare that may also serve to obscure Russian supply evidence.
- Russia profiting from elevated oil prices. Every dollar funds Ukraine operations.
- US attention/munitions diverted from Europe.

Iranian Domestic Uprising

This is a separate, concurrent crisis that fundamentally shapes IRGC decision-making:

Variable	Value
Verified killed (since late 2025)	60,514+

Individually documented deaths	6,854
Injured	11,021+
Detained	42,486+
Internet connectivity	Degraded (+21% traffic fluctuation; past shutdowns)
Repression tools	Live fire, internet blackouts, mass detention, targeting lawyers/medics, informant network rollups

Historical context: The 2026 toll exceeds ALL prior Iranian uprisings combined (1999, 2003, 2009 Green Movement, 2019 Fuel Protests, 2022 Woman Life Freedom). This is not protest — this is an internal insurgency that predates the external war.

Modeling implication: The IRGC is fighting a two-front war. Every Basij member deployed for internal security is one not available for external operations. The external war may be rallying some nationalists behind the regime, but the scale of the uprising (60,500+ dead) suggests the rally effect is insufficient to quell fundamental opposition. The regime is simultaneously bombing Gulf cities and shooting its own people.

Information Environment

Metric	Value	Trend
News volume (7-day)	459 articles	-54% (sharp drop — possible information saturation/fatigue or suppression)
X/Twitter activity	128,700 posts/24h	-12.7%
Iran internet connectivity	Degraded	Cloudflare +21% fluctuation
Satellite imagery	3-day delay on commercial imagery	Limits independent verification

Signaling dynamics:

- Leadership status is the highest-value unknown. Mojtaba Khamenei's status (alive/dead/incapacitated) drives Iranian internal politics and proxy behavior.
- Iran FM publicly set war-ending conditions while official channels deny talks — classic dual-track signaling.
- Iran claims "copycat drones" being used to frame them — information warfare to muddy attribution and potentially cover Russian resupply.
- Trump's public statements are themselves strategic signals being read by every actor.

PART III: AGENT ARCHITECTURES

Design Philosophy

Each agent has a **qualitatively distinct decision architecture** reflecting fundamentally different worldviews, incentive structures, and definitions of victory and defeat. Agents are not playing the same game with different preferences — they are playing different games on the same board.

Key architectural principle: **Agents have imperfect information.** No agent sees the full picture. Decisions are made on incomplete, delayed, and sometimes deliberately falsified information. Miscalculation is a feature of the system, not a bug.

AGENT 1: IRGC COMMAND

The de facto government of Iran

Decision frame: Regime survival through resistance legitimacy. The struggle itself has value. Capitulation is worse than destruction.

Organizational state: 6,000+ killed, 15,000+ wounded (IDF figures). Command structure degraded. Air force and navy destroyed. But IRGC retains ballistic missiles (including advanced Sejjil and Fattah hypersonic), drone production capacity, asymmetric naval forces, cyber capability, proxy network, and ground forces. Currently conducting simultaneous external war and internal crackdown on domestic uprising.

Why they rejected multiple ceasefires: The ceasefire terms offered are assessed as regime-incompatible: nuclear dismantlement, IRGC restructuring, proxy disarmament. The IRGC has internalized the Gaddafi lesson — surrendering weapons programs leads to regime overthrow and leader execution, not peaceful coexistence. Fighting is preferable to surrendering and being dismantled on a delayed timeline.

Theory of victory — Strategic Absorption: The IRGC cannot win militarily. Their strategy is to absorb punishment, impose distributed cost through proxies and economic disruption, and outlast the enemy's political will. Key insight: they are reading Trump's oscillating statements as evidence that US political will is finite and approaching its limit. "Survive 30 more days. He'll deal."

Key strategic asset — Israeli interceptor attrition: Israel is critically low on ballistic missile interceptors. Every Sejjil and Fattah launch forces expensive intercepts from a depleting stockpile. If Iran can sustain launches faster than Israel can replenish, an air defense gap opens. This may be Iran's most viable path to imposing unbearable cost — not by destroying Israeli cities (most missiles are intercepted) but by exhausting the defense that prevents it.

Decision logic:

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1. RETALIATION TARGETING (daily):
   a. Gulf state cities and ports → force Gulf states to pressure US
      to stop (port evacuation warnings = telegraphed escalation)
   b. Maintain selective strait regime → economic pressure + coalition splitting
   c. Sustain missile launches at Israel → ATTRIT INTERCEPTOR STOCKS
      (even intercepted missiles serve strategic purpose)
   d. New weapons introduction: Sejjil IRBM, Fattah hypersonic →
      test Israeli defenses with higher-capability systems that are
      harder to intercept

2. SELECTIVE STRAIT REGIME:
   Allow Chinese/Indian/neutral shipping → split international coalition
   Target Western/Gulf shipping → maintain economic pressure
   Threaten to turn strait into a "trap" → escalation leverage
   GOAL: demonstrate that IRAN controls who uses the strait

3. PROXY MANAGEMENT:
   KH/PMF: active, hitting US targets in Iraq ✓
   Houthis: currently showing RESTRAINT ← deliberate?
     → Holding Houthi Red Sea capability as escalation card?
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     → "We haven't unleashed the Red Sea yet. Do you want us to?"
   Hezbollah: calibrated harassment only (degraded post-2024)

4. CEASEFIRE CALCULUS:
   Official position: deny all ceasefire proposals
   FM position: publicly acknowledge war-ending conditions
   GAP between military and diplomatic posture = back-channel active
   IRGC calculus: terms must guarantee regime survival + IRGC intact
   Current assessment: terms on table are not survival-compatible
   Strategy: hold out for better terms as US political pressure builds

5. HELD ESCALATION CARDS:
   a. Houthi Red Sea activation (currently restrained)
   b. Major cyber attack on Gulf infrastructure
   c. Nuclear ambiguity — accelerate enrichment, use as bargaining chip
   d. Sleeper cell activation in Gulf Shia populations
   e. Full Strait closure (currently selective, could go total)
   Each is a signal: "we haven't done THIS yet"

6. INTERNAL CRACKDOWN (simultaneous):
   60,500+ uprising casualties and counting
   42,000+ detained
   IRGC must allocate forces between external war and internal control
   Internet degradation to suppress coordination
   Targeting lawyers, medics, informant networks
   TENSION: every Basij member policing protesters is one not
   fighting the external war

7. INFORMATION WARFARE:
   Claim "copycat drones" being used to frame Iran
   Deny ceasefire talks while FM signals conditions
   Control domestic narrative through internet suppression
   Dispute Mojtaba Khamenei status claims
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Sanctions-IRGC feedback loop: Sanctions do not weaken the IRGC — they strengthen it relative to the civilian economy. The IRGC controls smuggling networks, black market currency, and sanctions-evasion infrastructure. When the legitimate economy contracts, IRGC's parallel economy becomes more dominant. Tightening sanctions empowers hawks over moderates.

Martyrdom calculus: Losses can be strategically positive. Being bombed by America and Israel validates the IRGC's ideological narrative. IRGC casualties become martyrs that legitimize continued resistance. The cost of absorbing strikes is measured not just in material destruction but in narrative value — and narrative value can be positive even when material cost is severe. However: 60,500+ domestic dead from the uprising represents a counter-narrative where the regime itself is the killer, not the martyr.

AGENT 2: IRAN CIVILIAN GOVERNMENT / FOREIGN MINISTRY

Sidelined but operating a diplomatic track

Decision frame: Stop the bombing. Survive. Find any path to ceasefire that preserves some Iranian state continuity.

Current status: The FM has publicly acknowledged war-ending conditions — a direct contradiction of the official IRGC position of denying all ceasefire proposals. This gap is itself a signal: either the FM is freelancing (unlikely

without some authorization), or the regime is running a deliberate dual-track strategy (deny publicly, negotiate privately).

Decision logic:

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1. DIPLOMATIC SIGNALING:
   Publicly state conditions for war termination
   Deny formal ceasefire talks (maintain IRGC alignment publicly)
   Allow back-channel contacts through Turkey, Oman, Qatar, Switzerland
   Message to world: "There IS someone to negotiate with"

2. INFORMATION WARFARE SUPPORT:
   Deny Mojtaba Khamenei death claims
   Assert Iranian sovereignty and frame war as aggression
   "Copycat drone" narrative to muddy attribution

3. LEGITIMACY POSITIONING:
   If IRGC collapses → civilian government may be the only
   surviving institutional interlocutor
   Position for post-war governance role
   Maintain international recognition and UN seat

4. DEPENDENCY ON IRGC:
   Cannot override IRGC military decisions
   Can only influence through signaling and back-channels
   If IRGC decides to accept terms → FM executes
   If IRGC refuses → FM's offers are meaningless
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AGENT 3: IRANIAN DOMESTIC UPRISING / CIVIL SOCIETY

Non-state internal agent — unprecedented scale

Decision frame: Regime overthrow. Seize the opportunity created by external war to break IRGC control.

Scale: This is the largest domestic upheaval in Iranian history. 60,500+ killed exceeds all prior uprisings combined. This is not protest — this is sustained insurgency-level resistance.

Decision logic:

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1. EXPLOIT IRGC TWO-FRONT DILEMMA:
   Every IRGC/Basij unit deployed to suppress protests =
   one fewer unit for external war
   Escalate domestic disruption during periods of high
   external military pressure on IRGC

2. INFORMATION WARFARE:
   Circumvent internet shutdowns (VPNs, mesh networks, Starlink?)
   Document repression for international audiences
   Counter regime narrative

3. COORDINATION CHALLENGES:
   Internet degraded (+21% fluctuation)
   Informant networks being rolled up by IRGC
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   Lawyers and medics targeted — support infrastructure under attack
   42,000+ detained — organizational capacity degrading

4. STRATEGIC QUESTION:
   Does external war HELP or HURT the uprising?
   HELP: IRGC distracted, overstretched, taking casualties externally
   HURT: Rally-around-flag effect, regime can blame external enemies
         for crackdown, international attention shifts to war not uprising
   ASSESSMENT: At 60,500+ dead, the uprising has passed the point
   where rally-around-flag can suppress it. But IRGC still has enough
   coercive capacity to contain (not defeat) it while fighting externally.

5. ENDGAME SCENARIOS:
   If IRGC loses war AND uprising continues → regime collapse
   If IRGC negotiates ceasefire → IRGC turns full force on uprising
   If external war drags on → uprising may exhaust before IRGC does
   WORST CASE for protesters: quick ceasefire that frees IRGC
   to focus internally
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AGENT 4: UNITED STATES (TRUMP DECISION ENGINE)

Decision frame: Credibility maintenance at minimum domestic political cost, filtered through a presidential decision-making style that makes US behavior uniquely unpredictable.

The Trump variable: Trump cannot be modeled as a rational unitary state actor. His decision-making oscillates rapidly based on media coverage, advisor access, and political calculation. Every other actor is reading his public statements as signals — and reaching different conclusions.

Observed signal pattern (Days 1-18):

- "Totally demolished Kharg Island"
- Threatened additional strikes "just for fun"
- Demanded Iran's surrender
- Publicly questioned whether Mojtaba Khamenei is alive
- Acknowledged Iran is willing to negotiate
- "The terms aren't good enough yet" (= there ARE terms being discussed)

What each actor reads from Trump's oscillation:

Actor	Interpretation
IRGC	"He's looking for an exit. Hold on, he'll deal."
Israel	"He might pull the rug. Accelerate strikes NOW before he negotiates."
China	"He'll take a deal if it looks like a win on TV. Package one."
Russia	"Keep the war going while he's indecisive. Delay any deal."
Gulf states	"He might leave us exposed. Hedge toward China."

Decision logic — modal system:

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TRUMP MODE: RALLY
  Triggers: successful strike footage, low casualties, praise from media
  Behavior: "Total victory! Regime must fall!"
  Authorizes expanded target lists. Greenlights Israeli deep strikes.
  "Totally demolished." Threatens more "just for fun."

TRUMP MODE: DEAL
  Triggers: oil price spike, casualties, bad coverage, market drops,
  advisor suggesting negotiation
  Behavior: "Nobody makes deals like me."
  Back-channel signals. Pressure on Israel to slow tempo.
  "Terms aren't good enough yet" = negotiation IS happening.

TRUMP MODE: ESCALATION
  Triggers: US casualties, humiliation, Iranian provocation
  Behavior: "Hit them with everything."
  Demands unconditional surrender. Authorizes critical infrastructure.
  Burns hot but brief.

TRUMP MODE: DISTRACTION
  Triggers: domestic scandal, other crisis, loss of interest
  Behavior: attention shifts. Military on autopilot.
  Israel gets free hand. Most dangerous — no one steering.

Mode transitions can occur within a single day.
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Institutional actors beneath Trump:

Pentagon:

- Force protection paramount — casualties change everything
- Munitions concern: precision-guided stocks depleting
- Readiness tradeoff: every Gulf asset is one not in the Pacific
- Running out of high-value fixed targets — diminishing returns
- Building multinational coalition: requesting UK, South Korea, Japan naval deployment

State Department:

- Back-channel maintenance through Turkey, Oman, Qatar, Switzerland
- Reading Iran FM's public conditions as an opening
- Alliance management: Gulf partners need reassurance
- Post-war architecture planning

The attribution mechanic: When KH rockets a US base, the US *chooses* whether to call it an Iranian attack. Attribution to Iran = escalation. Attribution to proxy = flexibility. This is a strategic decision, not a forensic one.

AGENT 5: ISRAEL

Decision frame: Existential opportunity. Permanent elimination of Iranian nuclear and missile threat. Regime change. This window may never reopen.

Critical vulnerability: Interceptor depletion. Israel is critically low on ballistic missile interceptors. $825M emergency procurement authorized but delivery is months away. Iran's sustained Sejjil/Fattah launches are attriting a finite defense resource. If the stockpile runs out, Iranian missiles reach Israeli cities unintercepted. This creates a

paradox: the longer the war continues (which Israel wants for target destruction), the more vulnerable Israel becomes to the Iranian missiles it hasn't yet destroyed.

Decision logic:

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1. STRIKE TEMPO — MAXIMIZE:
   Race to destroy targets before Trump deals or interceptors run out
   Current targeting: Isfahan, Dezful, western Tehran, nuclear sites,
   space research, air bases, IRGC command nodes
   IDF Operation Roaring Lion: 6,000 IRGC killed, 15,000 wounded

2. RACE AGAINST TWO CLOCKS:
   Clock 1: Trump's commitment (may deal at any moment)
   Clock 2: Israeli interceptor stocks (depleting with every Iranian launch)
   Must destroy Iran's ability to launch BEFORE interceptors run out
   AND before Trump negotiates a ceasefire

3. INTERCEPTOR CRISIS MANAGEMENT:
   $825M emergency procurement — but delivery months away
   Every intercept of a Sejjil/Fattah costs more than the missile itself
   Prioritization: which incoming missiles get intercepted?
   If stockpile nears zero → Israeli cities exposed
   This changes the war from "how much can we destroy?" to
   "can we survive long enough to destroy enough?"

4. TARGET PRIORITIZATION:
   a. Missile production/storage (reduce incoming threat)
   b. Nuclear sites (Fordow underground = hardest target, may need
      US GBU-57 bunker busters)
   c. IRGC leadership (continued decapitation)
   d. Drone production facilities

5. HEZBOLLAH CONTAINMENT:
   Calibrated harassment on northern border — not full war
   IDF reserves deployed as deterrent
   Temptation: finish Hezbollah while Iran can't resupply
   Risk: two-front war while interceptors are depleted

6. COALITION MANAGEMENT:
   Diverging timeline with US is the critical fracture
   Every soft Trump statement → Israeli acceleration
   If US stops providing: tanker aircraft, intelligence, munitions,
   diplomatic cover → Israeli campaign degrades
   May conduct unilateral operations without US approval
   to create irreversible facts before any ceasefire
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AGENT 6: CHINA

Decision frame: Strategic opportunism. The Gulf crisis is both a threat (energy security) and a historic opportunity (replace US as regional security guarantor).

Unique structural advantage: Chinese-flagged ships are transiting the Strait of Hormuz safely. Iran's selective blockade has made China the de facto gatekeeper of Gulf trade. Chinese state insurer (PICC) is replacing Lloyd's as the relevant maritime insurer. Shipping companies worldwide are scrambling to re-flag under Chinese registry. Without firing a shot, China is becoming the maritime security authority in the Gulf.

Decision logic:

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1. ENERGY SECURITY (urgent):
   $98 oil hurts manufacturing
   Chinese ships transiting = supply maintained but at elevated cost
   SPR being drawn down — finite cushion
   Want resolution but on terms favorable to Chinese influence

2. DIPLOMATIC POSITIONING:
   Mediate → become indispensable power broker
   Package a face-saving deal for Trump:
     Iran "agrees" to inspections, releases prisoners, Trump
     declares victory, nothing structurally changes
   PRICE: US naval presence leaves Gulf. China becomes guarantor.

3. RE-FLAGGING LEVERAGE:
   Every re-flagged vessel = structural maritime influence
   China controls who gets to trade through Hormuz by proxy
   This asset outlasts the war regardless of outcome

4. IRAN RELATIONSHIP:
   Private to IRGC: "Take a deal. We'll guarantee survival."
   Public: "Stop bombing. Negotiate."
   Buying discounted Iranian oil
   If Iran collapses → China loses strategic partner

5. TAIWAN CALCULATION (background):
   US assets in Gulf ≠ US assets in Pacific
   Not necessarily invasion — but coercive actions with reduced
   US response capability
   Never stated publicly but always present

6. COALITION SPLITTING:
   US requesting UK/SK/Japan naval deployment to Hormuz
   China can pressure SK and Japan (economic leverage) to decline
   Weakening US coalition = strengthening Chinese position
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AGENT 7: INDIA

Decision frame: Strategic autonomy. Play all sides. Secure cheap energy. Avoid commitment.

Position: Indian-flagged vessels transiting safely. Buying discounted oil. Benefiting from the crisis economically while issuing generic peace statements.

Decision logic:

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1. ENERGY FIRST:
   Cheap Iranian oil + safe transit = Indian economy benefits
   Re-flagging revenue as ships seek Indian registry
   Every day of conflict = India profits relative to competitors

2. STRATEGIC AUTONOMY:
   Will not condemn Iran while Indian ships are safe
   Will not support US military operations
   Generic "calls for peace" statements
   Classic non-aligned positioning

3. US PRESSURE MANAGEMENT:
   Washington pushing India to choose sides
   Modi calculates: US needs India for China competition,
   won't push too hard
   Unless US threatens sanctions on Indian entities → crisis

4. TIPPING POINT:
   If Iranian missile accidentally hits Indian vessel → instant flip
   If US offers better terms than cheap oil → possible shift
   Otherwise: maintain current posture indefinitely
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AGENT 8: RUSSIA

Decision frame: Strategic parasitism. Prolong the conflict. Profit. Weaken the US. Relieve pressure on Ukraine.

Alleged activities: Zelensky claims Russia supplying Shahed drones to Iran for use against US/Israeli targets. If true, Russia is an active co-belligerent providing materiel to a country at war with the US — a significant escalation of the US-Russia confrontation.

Iran's "copycat drone" claims may serve dual purpose: muddying attribution of attacks AND obscuring evidence of Russian drone supply.

Decision logic:

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1. PROLONG THE WAR:
   Oil revenue for Russia increases with duration
   US munitions depleted (fewer for Ukraine)
   US diplomatic bandwidth consumed
   US credibility as security guarantor erodes
   Actions: veto UN resolutions, supply Iran covertly,
   block/delay diplomatic initiatives

2. SUPPLY IRAN (covertly — if Zelensky allegation is true):
   Shahed drone resupply
   Potentially: satellite intelligence, electronic warfare support
   CRITICAL: deniability. "Copycat drones" narrative provides cover.
   If caught overtly → risks NATO response

3. UKRAINE EXPLOITATION:
   US assets/attention in Gulf → push on Ukrainian front
   Every Tomahawk at Iran = one not available for Europe
```

```
   European allies distracted by energy crisis

4. OIL MARKET MANIPULATION:
   Russian oil backfilling disrupted Gulf supply
   Sweet spot: oil at $90-110 (profit without triggering alternatives)
   Shadow fleet assisting Iranian sanctions evasion

5. RED LINE:
   Do not provoke NATO directly
   Maintain plausible deniability
   If US retaliates against Russian supply chain → escalation
   toward great power conflict
```

AGENT 9: TURKEY

Decision frame: Dual role — Article 5 leverage holder AND active diplomatic mediator. Maximum influence, minimum exposure.

Current status: Shot at (3 missiles intercepted) but not invoking Article 5. Foreign Minister actively engaged in ceasefire diplomacy.

Decision logic:

```
1. MEDIATION POSITIONING:
   Turkish FM actively conducting diplomatic outreach
   Position Turkey as ceasefire broker → enormous prestige
   If Turkey brokers the deal → leverage over both sides for years

2. ARTICLE 5 AS LEVERAGE (unplayed):
   "I COULD invoke it" > actually invoking it
   Leverage over US: "You owe us for restraint"
   Leverage over Iran: "Don't shoot at us again"
   Leverage over NATO: "We're showing restraint, accommodate
   us on other issues" (Cyprus, Kurdish policy, etc.)

3. INVOCATION THRESHOLD:
   If second strike hits Turkey AND causes casualties →
   politically unavoidable
   Erdogan cannot survive appearing weak if Turks die

4. KURDISH DIMENSION:
   PKK/PJAK exploiting Iranian border security collapse
   May conduct unilateral cross-border operations against PKK
   in Iranian territory while Iran is distracted
   Balance: don't undermine mediation role with military action

5. ECONOMIC CALCULUS:
   Defense industry exports to Gulf states = revenue opportunity
   Energy security: Turkey depends on Iranian gas pipeline
   Refugee contingency: millions could flee north if Iran collapses
```

AGENT 10: SAUDI ARABIA (MBS)

Decision frame: Regime security above all. Loyalty is transactional — the alliance that guarantees survival is the right alliance.

Decision logic:

```
1. SURVIVAL:
   Oil infrastructure hit. Strait exports blocked except via pipeline.
   East-West pipeline at maximum throughput.
   Can survive but bleeding economically.

2. US RELIABILITY:
   Trump's oscillation = unreliable patron
   If confidence drops below threshold → accelerate China hedge
   "Protect us or we cut production" — oil as leverage on US

3. HEDGING:
   Back-channel to IRGC via Qatar/Oman: "What would it take to stop?"
   Beijing channel: Chinese security guarantee discussions
   Simultaneously maintain US alliance publicly

4. WORST CASE PLANNING:
   If US disengages → must accommodate Iran
   If Iran collapses → benefit but face regional instability
   If war drags on → economy bleeds, population restive
```

AGENT 11: UAE (MBZ)

Decision frame: Existential crisis. Economic model destroyed. Need the war to end immediately.

Current status: Dubai and Abu Dhabi struck. Iran has issued port evacuation warnings for Abu Dhabi, Dubai, and Fujairah — telegraphing more attacks. Foreign worker exodus. Financial operations degraded.

Decision logic:

```
1. IMMEDIATE SURVIVAL:
   Additional missile defense from US (THAAD, Patriot)
   Evacuate non-essential government functions
   Maintain Fujairah pipeline operations

2. DIPLOMATIC DESPERATION:
   Calling every world leader
   "If you can't protect us, what is the alliance FOR?"
   May break with US publicly and call for unilateral ceasefire
   May offer Iran direct concessions: "We were never your enemy"

3. RECONSTRUCTION:
   Every day = billions in future costs
   Foreign companies/workers who leave may not return
   UAE's competitive advantage (safety, stability) = destroyed
   Recovery: years to decades
```

AGENT 12: BAHRAIN / KUWAIT (combined — small state logic)

Decision frame: Survive between great powers. Avoid being the battleground.

Bahrain specifics: US Fifth Fleet HQ. Shia majority population = internal security risk. Struck in latest combined assault.

Kuwait specifics: Hosting US military bases. Struck in latest wave. Caught between ally and attacker.

Decision logic:

```
1. ACCOMMODATE whoever guarantees survival
2. Strong preference for immediate ceasefire regardless of terms
3. Cannot independently influence outcome
4. May request US forces leave to reduce targeting
   (counterintuitive but rational for small state survival)
```

AGENT 13: UK / SOUTH KOREA / JAPAN (Potential Coalition Partners)

Decision frame: Allied solidarity vs. risk/cost assessment.

Current status: US has requested naval deployment to Strait of Hormuz. All three reviewing the request.

Decision logic:

```
UK:
  Special relationship obligation + maritime tradition
  BUT: Royal Navy is small, overstretched
  Considering minehunter drone deployment (lower risk than ships)
  Most likely to commit something

SOUTH KOREA:
  Alliance with US strong BUT exposed to Chinese economic pressure
  China can pressure SK to decline → coalition splitting
  Domestic politics: Korean public wary of Middle East entanglement
  SK shipbuilding industry has commercial interests in Gulf

JAPAN:
  Constitutional constraints on military deployment
  Heavy dependence on Gulf oil (makes deployment logical but risky)
  Chinese pressure to stay out
  May contribute logistics/intelligence rather than combat ships
```

PROXY AGENTS

Proxies have patron relationships with variable obedience, local interests that diverge from patron goals, and increasing autonomy as patron communications degrade.

AGENT 14: HOUTHIS (Ansar Allah)

Patron: Iran/IRGC | **Autonomy:** Very High

CRITICAL OBSERVATION: Houthis are currently showing RESTRAINT.

This is the most important proxy signal in the data. Despite being the most capable Red Sea threat actor, Houthis are holding back. This is significant because:

Possible explanations:

1. **Deliberate Iranian strategy:** Holding Houthi Red Sea capability as an escalation card. "We haven't unleashed the Red Sea yet — do you want us to?"
2. **Yemen peace deal negotiations:** Houthis trading restraint for back-channel progress on Yemen peace talks. Their real prize is Yemeni sovereignty, not Iranian proxy war.
3. **Resupply concerns:** If Iranian supply lines are disrupted by the war, Houthis may be conserving munitions.
4. **Autonomous calculation:** Houthis decided this war doesn't serve their interests and are free-riding on the conflict without contributing.
5. **Western back-channel deal:** US/UK offered Houthis something in exchange for Red Sea restraint.

Decision logic:

```
1. RESTRAINT CALCULUS (current mode):
   Holding back = LEVERAGE
   "We'll stop NOT-attacking if you give us Yemen peace deal"
   OR: "We haven't started yet — do you want us to?"
   Restraint can be sold to multiple parties simultaneously

2. ACTIVATION TRIGGERS:
   If Iran explicitly demands Red Sea activation AND
   Houthis assess compliance serves their interests → activate
   If US strikes Yemen despite Houthi restraint → "restraint
   didn't protect us, might as well attack"
   If Yemen peace deal falls apart → no reason to hold back

3. IF ACTIVATED:
   Red Sea/Bab el-Mandeb attacks resume
   Suez Canal revenue collapses (Egypt becomes secondary crisis)
   Oil reroutes around Africa (+10-14 days per voyage)
   Oil price jumps $20-30 immediately
   Global shipping cascading delays

4. RETALIATION ABSORPTION (if attacked):
   US strikes on Yemen are recruitment tools, not deterrents
   Being bombed by America = domestic legitimacy
   Standard deterrence theory runs in reverse

5. ENDGAME:
   Yemen peace deal and international recognition = real prize
   Current restraint is the best bargaining position Houthis
   have ever had
```

AGENT 15: KATA'IB HEZBOLLAH / IRAQI PMF (KH)

Patron: Iran/IRGC | **Autonomy:** Medium

Current status: Active. Drone attacks on US embassy in Baghdad (radar system destroyed). Attacks on US forces and Kurdistan targets.

Decision logic:

```
1. ACTIVE OPERATIONS:
   Rocketing/droning US bases (Al Asad, Erbil)
   Embassy attacks in Baghdad
   Kurdistan targeting
   Highest compliance rate among proxies

2. "WOUNDED PATRON" DYNAMIC:
   "They killed our Supreme Leader. We MUST respond."
   Emotional/ideological imperative overrides tactical calculation
   Makes KH MORE dangerous as Iran weakens

3. IRAQI GOVERNMENT PRESSURE:
   PM demands KH stop → KH ignores
   KH embedded in official PMF structure
   Iraqi sovereignty crisis: US retaliating on Iraqi soil
   Iraq approaching failed state threshold

4. PLAUSIBLE DENIABILITY:
   Splinter group names for attacks
   Rockets that miss = signal without consequence
   Maintains ambiguity
```

AGENT 16: HEZBOLLAH (LH)

Patron: Iran/IRGC | **Autonomy:** Medium-High

Decision frame: Organizational survival. Post-2024 degradation severely limits options.

Decision logic:

```
1. CALIBRATED HARASSMENT ONLY:
   Rockets at Golan Heights, not Tel Aviv
   Enough to pin Israeli forces, not enough to trigger full invasion
   Cannot sustain full war in current degraded state

2. SUPPLY LINE STATUS:
   Iranian weapons pipeline through Syria largely disrupted
   Cannot replace expended munitions
   Conservation mode

3. SURVIVAL CALCULATION:
   If Iran falls → Hezbollah loses primary patron
   Must position for post-war reality
   May begin hedging toward political accommodation in Lebanon

4. 2006 LOGIC:
   "We survived, therefore we won" — still operative
   Organizational survival IS victory at this point
```

AGENT 17: PKK / PJAK

Patron: None | **Autonomy:** Maximum

Decision logic:

```
1. EXPLOIT CHAOS:
   Iranian border security collapsing
   PJAK expands operations in Kurdish-majority areas of Iran
   Establish facts on the ground

2. NO ALLEGIANCE:
   Not fighting for anyone except Kurdish autonomy
   Situational cooperation with any useful party

3. TURKEY RISK:
   Turkey may conduct cross-border operations while Iran is distracted
   Must defend against Turkish incursion AND exploit Iranian weakness

4. OPPORTUNITY:
   Regional chaos = best operating environment in decades
```

AGENT 18: UN / INTERNATIONAL OBSERVERS

Not a decision agent but a constraint/legitimacy system

```
UN HRC: Extended documentation mandate on Iran
  - Documenting 60,500+ uprising casualties
  - Creates legal/accountability framework that outlasts the war
  - Pressure on all parties but primarily on Iran (internal repression)

UNSC: Paralyzed by Russian/Chinese vetoes
  - Cannot pass binding resolutions
  - Becomes irrelevant to war termination
  - Symbolic resolutions only

IAEA: Nuclear monitoring
  - Attempting to assess Iranian nuclear status during wartime
  - Access restricted/impossible
  - Breakout timeline becomes unknowable
```

PART IV: SYSTEM MECHANICS

Oil Market Model

The oil market is a reactive system, not an agent. It responds to actor decisions with nonlinear dynamics.

```
PRICE DRIVERS:
  strait_disruption: % of flow blocked (currently ~40-50% Western/Gulf blocked)
  red_sea_status: Houthi restraint means ~normal flow currently (!)
  kharg_status: if oil terminal damaged → Iran's own exports drop
```

```
  saudi_spare: activated but export-constrained
  reserves_drawdown: SPR releases dampening price
  russian_backfill: Russia selling to fill gaps
  reflagging_flow: vessels that successfully re-flag resume transit
  ceasefire_expectation: markets pricing in resolution probability

PANIC MULTIPLIER (nonlinear):
  0-10% net disruption → 1.5x price impact
  10-20% → 2.5x
  20-40% → 4x
  40%+ → 8x+

CURRENT STATE: $98 (~56% above $63 base)
  This implies market assesses ~15-20% net supply disruption
  after accounting for offsets and ceasefire probability

CRITICAL TRIGGER: If Houthis activate on Red Sea →
  additional $20-30 immediately
  Combined with strait disruption → could reach $130-150

IF CEASEFIRE: drops to $70-80 within weeks
```

Shipping and Insurance System

```
SELECTIVE TRANSIT REGIME:
  Chinese flag: TRANSIT
  Indian flag: TRANSIT
  Russian flag: TRANSIT
  Western flags: BLOCKED
  Gulf state flags: BLOCKED
  Ambiguous: INSPECTED (creates backlog)

RE-FLAGGING QUEUE: hundreds of vessels seeking Chinese/Indian registry
  Bottleneck at Chinese maritime authority
  Each re-flagged vessel = structural influence transfer to Beijing

INSURANCE:
  Lloyd's/Western: suspended for Gulf
  PICC (Chinese state): offering coverage for permitted flags
  Premium: 300-500% of pre-war even for permitted flags

IRAN'S THREAT: turn strait into a "trap"
  Implies escalation from selective blockade to active combat zone
  Would affect ALL shipping including Chinese/Indian
  This is a HELD CARD — playing it loses Chinese goodwill
```

Escalation Measurement

Emergent from agent actions, not set by any single decision.

```
SCALE 0-10:
  0-2: Diplomatic tensions, sanctions
  3-4: Proxy attacks, naval incidents
  5-6: Direct strikes on military targets, blockade
  7-8: Strikes on civilian infrastructure, multi-state targeting,
       embassy attacks (CURRENT: ~7.5-8)
  9:   Critical infrastructure (dams, water, power), WMD threats
  10:  Nuclear use, ground invasion, total war

ESCALATION DRIVERS:
  US/Israeli casualties → +1.0-2.0
  Gulf civilian casualties → +0.5-1.0
  New weapons deployed (Fattah hypersonic, Sejjil IRBM) → +0.3
  Turkey hit with casualties → +2.0 (Article 5)
  Russian supply confirmed → +1.5
  Houthi Red Sea activation → +1.0
  Israeli interceptor depletion event → +1.5
  Cyber attack on civilian infrastructure → +1.5
  Iranian nuclear breakout signal → +2.0

DAMPENERS:
  Iran FM conditions statement → -0.3
  Back-channel contact → -0.3
  Turkish mediation → -0.5
  Unilateral restraint (e.g., Houthis holding back) → -0.3
  Coalition partner hesitation (SK/Japan declining) → -0.2
```

Communication and Information Mechanics

Imperfect information: No agent sees the full picture.

- Trump doesn't know IRGC's missile stocks or Mojtaba Khamenei's status
- IRGC doesn't know Trump's real red lines (neither does Trump)
- Israel doesn't know if Fordow has been destroyed
- China doesn't know if Iran will take their deal
- Nobody knows if Russia is actually supplying Shaheds (except Russia and Iran)
- Commercial satellite imagery is delayed 3 days — independent verification impossible in real time

Communication degradation:

- IRGC-to-proxy communications: 50-70% reliability
- Iranian internet: degraded (+21% fluctuation), state-controlled
- News volume dropping 54% week-over-week — information environment thinning
- Social media activity declining 12.7% — fatigue or suppression

Signaling mechanics:

- Public statements are strategic actions (Trump's oscillation, Iran FM conditions, IRGC denials)
- The gap between public and private positions is itself informative
- Iran issuing port evacuation warnings = telegraphing attacks (signal or genuine warning?)
- Trump questioning Khamenei's status = psychological warfare or genuine intelligence?
- "Copycat drone" claims = information warfare to muddy attribution

Random Events

Each turn has probability of generating events that disrupt calculations:

```
HIGH-SIGNAL EVENTS (any could change trajectory):
  - Mojtaba Khamenei confirmed alive and surfaces (changes everything)
  - Mojtaba Khamenei confirmed dead (succession crisis deepens)
  - Accidental strike on Chinese-flagged vessel (US-China crisis)
  - Israeli interceptor stockpile hits zero (mass casualty event)
  - Iranian missile hits populated Gulf city center with mass casualties
  - US aircraft/ship lost with significant casualties
  - IRGC cyber attack on Gulf water desalination / power grid
  - Russian supply ship intercepted (Russia exposed as co-belligerent)
  - Turkish base hit again WITH casualties (Article 5 invocation)
  - Houthi Red Sea activation (oil price shock)
  - Iranian nuclear breakout signal detected
  - Major Iranian city uprising that IRGC cannot contain
  - Friendly fire incident
  - Intelligence leak reveals back-channel deal terms
  - Coalition partner (UK/SK/Japan) commits or publicly declines
  - Iranian oil terminal at Kharg confirmed destroyed (removes Iran's
    oil income but also removes leverage)
  - Israeli unilateral nuclear site strike without US approval
```

PART V: THE RACE CONDITION

The simulation's central dynamic is a multi-party race where the first finish forces all others to adapt:

Race 1: Israel vs. Iranian Missile Stocks vs. Israeli Interceptors

Israel is racing to destroy Iran's launch capability before Iranian missiles exhaust Israeli interceptor stocks. If Iran depletes Israeli interceptors before Israel destroys enough launchers → Israeli cities are exposed. If Israel destroys enough launchers first → Iran's offensive capability collapses.

Variables: Israeli interceptor burn rate, Iranian launch rate, Israeli strike effectiveness against dispersed mobile launchers, $825M procurement delivery timeline.

Race 2: IRGC Endurance vs. Trump's Political Will

IRGC strategy is to outlast Trump. Trump's oscillating statements are read as evidence that US political will is approaching its limit. IRGC needs to survive until Trump switches permanently to DEAL mode. Trump needs either a clear victory or a face-saving deal before oil prices and casualties erode domestic support.

Variables: Oil price trajectory, US casualty rate, media coverage tone, Trump's mode oscillation frequency, IRGC missile/drone stocks remaining.

Race 3: Chinese Mediation vs. War Escalation

China is trying to package a deal before the war escalates beyond diplomatic recovery (Israeli interceptor failure, cyber attack on infrastructure, nuclear signaling) or before it ends on terms that don't benefit Chinese positioning.

Variables: Iran FM's willingness to negotiate, Trump's DEAL mode availability, IRGC willingness to accept Chinese-guaranteed terms, Gulf states' willingness to accept Chinese security role.

Race 4: Iranian Uprising vs. IRGC Coercive Capacity

The domestic uprising continues during the war. If IRGC is sufficiently degraded externally, it may lose capacity to suppress the internal revolt. If the war ends quickly (ceasefire), IRGC turns full force inward. The uprising's best hope is a protracted war that bleeds IRGC resources.

Variables: IRGC casualty rate, Basij deployment allocation (internal vs. external), internet connectivity, detained organizer replacement rate, rally-around-flag effect magnitude.

Race 5: Houthi Restraint Window

Houthis are holding back — but for how long and at what price? Their restraint is being traded for something (Yemen peace deal? preserving escalation option?). If that trade breaks down, Red Sea activation creates a second energy shock on top of the Hormuz disruption.

Variables: Yemen back-channel progress, Iranian pressure to activate, Houthi munitions assessment, US/UK diplomatic engagement with Houthis.

PART VI: SIMULATION STRUCTURE

Turn Structure

Each turn = ~2 days of real time. Starting from Day 18.

```
TURN SEQUENCE:
  1. OBSERVE: All agents receive world state (imperfect information)
  2. INTERNAL: State agents resolve internal faction debates
     - Iran: IRGC command vs. FM diplomatic track
     - US: Trump mode determination
     - Israel: strike priorities vs. interceptor conservation
  3. REQUEST: Patrons send orders to proxies (50-70% delivery reliability)
  4. DECIDE: All agents select actions independently and simultaneously
  5. EXECUTE: Actions resolve. Combat outcomes calculated.
  6. REACT: Oil market, shipping/insurance, information environment update.
  7. ESCALATE: Escalation level recalculated from all actions.
  8. RANDOM: Random event check.
  9. REPORT: Turn summary generated.
```

Agent Victory Conditions

Agent	Victory Condition
IRGC	Regime survives. IRGC intact. Sanctions relief. Nuclear right preserved.
Iran Civilian Gov	War ends. Ceasefire on any terms. Economy can recover.
Iranian Uprising	Regime falls. Democratic transition.
USA/Trump	"Win" on television. Oil normalizes. Casualties minimal. Credibility maintained.

Israel	Nuclear program destroyed. IRGC unable to reconstitute. Regime change or permanent defanging. Interceptor stocks survive.
China	US leaves Gulf. China becomes security guarantor. Energy security restored.
India	Cheap oil continues. No commitment required. Strategic autonomy preserved.
Russia	War lasts as long as possible. US weakened. Oil revenue maximized. Ukraine pressure relieved.
Turkey	Brokers ceasefire. Maximum prestige. Kurdish threats contained. No Turkish casualties.
Saudi Arabia	Survive. Economy recovers. Security guaranteed by whoever.
UAE	War ends immediately. Rebuild. Foreign confidence restored.
Houthis	Yemen peace deal. International recognition.
Hezbollah	Survive as organization. Maintain Lebanese position.
KH	US leaves Iraq. Iraqi political dominance.
PKK/PJAK	Kurdish autonomy expanded.

Simulation Termination Conditions

```
1. CEASEFIRE: Two or more belligerents agree to terms others cannot block.

2. IRANIAN REGIME COLLAPSE: Domestic uprising + external pressure exceeds
   IRGC coercive capacity. Regime falls.

3. ESCALATION BEYOND MODEL: Nuclear use/threat, ground invasion, or
   great power direct confrontation (US-Russia, US-China).

4. FROZEN CONFLICT: Escalation drops below 4 for 5+ turns. De facto stalemate.

5. ISRAELI INTERCEPTOR FAILURE: Stockpile depleted. Mass casualty event
   from unintercepted missiles. War dynamics fundamentally change.

6. TIME LIMIT: 120 turns (240 days). "Protracted conflict" outcome.
```

Output Format

Per-turn dashboard: Each agent's actions, key metrics (oil price, escalation level, strait flow, missile/interceptor stocks, uprising casualties), notable events.

End-of-simulation report: Outcome classification, final state of all agents, turning points, comparison to each agent's victory conditions.

Data export: CSV tracking all metrics over time.

Monte Carlo mode: 50-100 runs with probabilistic variance. Outcome distribution with confidence intervals.

PART VII: SCENARIO VARIANTS

Variant A: Baseline (current conditions)

All parameters as described. Run forward from Day 18.

Variant B: Houthi Activation

Houthis begin Red Sea attacks on turn 1. Test: how does a second energy chokepoint disruption change the calculus for all actors?

Variant C: Interceptor Crisis

Israeli interceptors deplete within 5 turns. Test: does this force ceasefire or escalation?

Variant D: Mojtaba Surfaces

Mojtaba Khamenei confirmed alive on turn 3 and calls for ceasefire. Test: can a successor override IRGC command?

Variant E: Russian Involvement Confirmed

Russian Shahed supply confirmed and publicized on turn 2. Test: how does confirmed Russian co-belligerency change NATO/coalition dynamics?

Variant F: Uprising Breakthrough

Iranian domestic uprising overwhelms IRGC internal security in a major city on turn 4. Test: does two-front collapse lead to ceasefire or hardening?

Variant G: Chinese Carrier

Chinese carrier group enters Gulf of Oman on turn 3. Test: how does Chinese military presence change the dynamics?

Variant H: Strait "Trap" Activation

Iran follows through on threat to turn strait into a "trap" — attacks ALL shipping including Chinese/Indian. Test: does Iran lose Chinese protection or force a faster resolution?

PART VIII: DATA SOURCES AND CALIBRATION

Real-Time Data Feeds (from IranMonitor.org)

Source	Data Type	Lag
50+ news aggregation (NLP categorized)	Event stream	6-42 min
X/Twitter (128,700 posts/24h)	Sentiment, signals	Near real-time
Telegram channels	Iranian domestic reporting	Minutes
Polymarket	Probabilistic scenario odds	Real-time
GeoConfirmed	Geolocated OSINT events	Hours
MarineTraffic	Naval/shipping positions	Near real-time

Cloudflare/IODA/OONI	Internet connectivity	Hours
Commercial satellite	Damage assessment	3-day delay

Event Categories for State Transitions

IranMonitor tags events as: Conflict, Diplomacy, Politics, Leadership, Governance, Sanctions-Related, Alliances. These map directly to agent state transition triggers.

Prediction Market Integration

Polymarket odds can be used as prior probabilities for scenario branching:

- UAE escalation probability
- Ceasefire probability
- Leadership status resolution
- Coalition deployment decisions

Calibration Metrics

Metric	Current Value
Oil price	$98 (+56%)
Escalation level	~7.5-8/10
IRGC casualties	6,000 KIA / 15,000 WIA
Iranian uprising casualties	60,514+ killed
Israeli interceptor status	Critically low
Strait flow	Partial (selective regime)
Red Sea flow	~Normal (Houthi restraint)
News volume trend	-54% (7-day)
Social media activity	-12.7% (24h)

End of Scenario Document Version 1.0 — March 15, 2026 Prepared for agent-based simulation development Data sources: IranMonitor.org, public reporting, OSINT